

Lecture 3 Reporting Verbs

♣ Reporting Verbs Practice

Use the verbs below to complete the sentences by reporting research findings.

argue	claim	emphasize	explain	suggest
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- Harris (2021) _____ that Bitcoin has no value whatsoever. *(says indirectly that something is true)*
- Samuel and Preston (2021) _____ that the increase spending for councils in England is not adequate enough to address the issues in the care sector. *(they give reasons for their view)*
- Watson (2021) _____ that he has had a major breakthrough in the fight against cancer. *(says something is true directly, and firmly, often used when others disagree).*
- Asprey (2018) _____ that exercise is a powerful anti-aging tool. *(to highlight an important point).*
- Banks (2021) _____ how the UK government plans to launch a rocket into orbit next year. *(give clear details about something)*

♣ Synthesis Practice

Highlight the sources by identifying the reporting verbs and the corresponding points made.

6. *Coursework Versus Examination Assessment*

Using assignment essays for assessment supports learning better than the traditional examination system. It is considered that course-work assignment essays can lessen the extreme stress experienced by some students over “sudden death” end of semester examinations and reduce the failure rate (Langdon, 2016). Study skills research by Peters et al. (2018) support assessment by assignment because research assignments can be used to assess student learning mid-course and so provide them with helpful feedback. They also consider that assignment work lends itself to more critical approaches which help the students to learn the discourse of their subjects. In contrast, Abbot (2008) and Cane (2018) both argue that assignments are inefficient, costly to manage and are the cause of plagiarism problems in universities. A key argument is that “assessment by examination is a clean-cut approach as you obtain students’ knowledge under supervised circumstances” (Bable, 2008, p.20). The weight of evidence, however, would suggest that it is a fairer and more balanced approach to have some assessment by assignment rather than completely by examinations.

7. *Surmounting the Ceiling Effect of Motor Expertise by Novel Sensory Experience with a Hand Exoskeleton*

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(Quoted from: Furuya et al., Surmounting the ceiling effect of motor expertise by novel sensory experience with a hand exoskeleton. Science Robotics, eadn3802, 2025.)

EXOSKELETONS

Surmounting the ceiling effect of motor expertise by novel sensory experience with a hand exoskeleton

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For trained individuals such as athletes and musicians, learning often plateaus after extensive training, known as the “ceiling effect.” One bottleneck to overcome it is having no prior physical experience with the skill to be learned. Here, we challenge this issue by exposing expert pianists to fast and complex finger movements that cannot be performed voluntarily, using a hand exoskeleton robot that can move individual fingers quickly and independently. Although the skill of moving the fingers quickly plateaued through weeks of piano practice, passive exposure to otherwise impossible complex finger movements generated by the exoskeleton robot at a speed faster than the pianists’ fastest one enabled them to play faster. Neither a training for fast but simple finger movements nor one for slow but complex movements with the exoskeleton enhanced the overtrained motor skill. The exoskeleton training with one hand also improved the motor skill of the untrained contralateral hand, demonstrating the intermanual transfer effect. The training altered patterns of coordinated activities across multiple finger muscles during piano playing but not in general motor and somatosensory functions or in anatomical characteristics of the hand (range of motion). Patterns of the multifinger movements evoked by transcranial magnetic stimulation over the left motor cortex were also changed through passive exposure to fast and complex finger movements, which accompanied increased involvement of constituent movement elements characterizing the individuated finger movements. The results demonstrate evidence that somatosensory exposure to an unexperienced motor skill allows surmounting of the ceiling effect in a task-specific but effector-independent manner.

INTRODUCTION

Mastery of the outstanding skills characterizing experts accompanies extensive practicing. However, practice does not always guarantee improvement. Recently, an increasing number of studies have demonstrated that the amount of training accounts for less than half of mastery of expertise, which questions the impacts of deliberate practice (1–4). This is particularly important for trained individuals who seek to develop further sophisticated skills but face a challenge in achieving it because of a diminished learning gain after extensive training for stabilizing the skill, the so-called ceiling effect (5). To surmount limits of expertise, it is essential to establish unique ways of training that do not rely only on the amount of training or early training in childhood (6–8). Recent studies have reported the effectiveness of specialized sensorimotor training for well-trained individuals (9, 10). However, most of these studies have focused on simple motor tasks (e.g., repetitive motions with a single finger). It remains elusive what training strategies enhance the complex motor skills of trained individuals, which limits both a comprehensive understanding of the neuroplastic mechanisms of the sensorimotor system and the development of training methods that enable the enhancement of overtrained skills.

A common dilemma in further sophisticating complex motor skills that have undergone extensive training is the absence of prior physical experience with the extraordinary skill to be acquired. This can make it challenging for learners to obtain precise error information between the desired and produced body movements because of imprecision of motor simulation and/or motor imagery of the unexperienced complex skill (11–14). Thus, movement reorganization for further sophisticating trained motor skills can be achieved exploratorily for minimizing

performance errors and/or maximizing reward (15, 16). One potential method to optimize this learning is to provide sensory information on the target motor skill (17), such as vision-based observation learning (18, 19) and somatosensory learning through exposure to passively generated movements (20, 21). For example, limb motions passively generated by a robot facilitated accuracy of the limb movements and postural control in healthy individuals and patients with movement disorders (20–23). One may therefore postulate that exposure to sensory stimuli representing complex motor skills such as dexterous finger maneuvers similarly further improves overtrained skills. However, most previous studies that addressed the effects of passive sensory experience on motor skills have investigated simple tasks, which include reaching, grasping, walking, and standing (24–26). Several studies also reported learning of complex motor skills such as Braille reading, typing, and piano playing through haptic devices by nonexperts (27–29). Whether exposure to previously unexperienced sensory experiences of a complex motor skill can further improve the well-trained skill and thereby overcome the ceiling effect has not been addressed.

To test this, we used a custom-made exoskeleton robot for the hand that can move the individual fingers independently for both flexion and extension. Passive training effects with the robot were predicted on the basis of the observation that tuning movement patterns of overlearned skills such as standing and running by attaching an exoskeleton robot to the limb enhanced motor performance (30–34); however, it remains unknown whether the enhancing effect is limited only when the robot is attached to the limb. To identify what sensory experience can enhance a complex overtrained motor skill, the exoskeleton robot was used by expert pianists to experience different patterns of multifinger movements at different speeds. We postulated that somatosensory inputs of unexperienced motions, which involved faster and more accurate performance of complex multifinger movements than voluntarily produced motion, specifically enhance the motor skill of trained pianists, even after the skill

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➤ Exercise

Synthesize the information in the table using proper reporting verbs. Remember to use accurate referencing (integral or non-integral) and the correct transition words.

Johnson (2018)	Williams (2019)	Prompt
The main goal of the World Bank is to reduce poverty and foster economic growth in developing countries.	There has been an increase in the level of poverty in Africa.	Counter-argue.

Arnold (2019)	James (2020)	Prompt
The decline of printed newspaper is due to increased online activity overall.	The more time we spend online, the more likely we are to use search engines and social media for news.	Build on former view.
